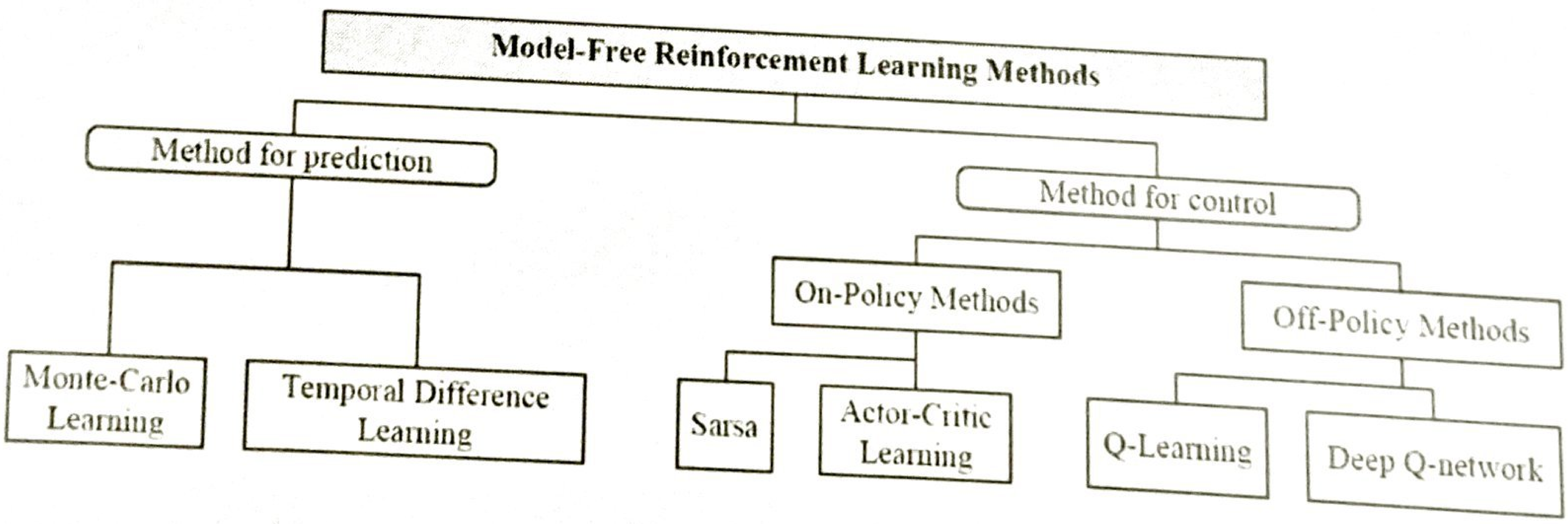
Model free learning

After the basic introduction of RL and its main framework (MDP), we now provide an overview of several commonly used RL methods. As a matter of fact, most traditional MDP problems assume that the full knowledge of environment is known beforehand. Hence, we can use dynamic programming (DP) methods such as policy iteration (PI) and value iteration (VI) to solve these MDP problems. Nonetheless, RL problems focus on dynamic environment, in which the environment model is unknown, i.e., both transition matrix and reward functions are unknown/inaccurate. Therefore, several RL methods including *model-based* methods and *model-free* methods are proposed. The former learns to fit the environment model, transforming RL problems into traditional DP problems, while the latter is like trial-and-error learning, drawing much attention.

Thus, we particularly discuss *model-free methods* here. Especially, these methods can be further categorized into two classes, i.e., *methods for prediction* and *methods for control*. The former is to estimate the *value function* for an underlying specific policy, while the latter is to learn *optimal value function* as well as optimal policy together. Moreover, depending on whether the optimal action-value function is updated by the policy being followed, methods for control are divided into two categories, i.e., on policy and off-policy methods. The former estimates state value function assuming the current policy continues being followed, while the latter estimates it assuming a greedy policy is followed rather than the current policy. Following is the taxonomy of model-free learning approaches:



Ref: Reinforcement Learning Meets Wireless Networks: A Layering perspective
Published in IEEE IoT

Temporal-difference Learning

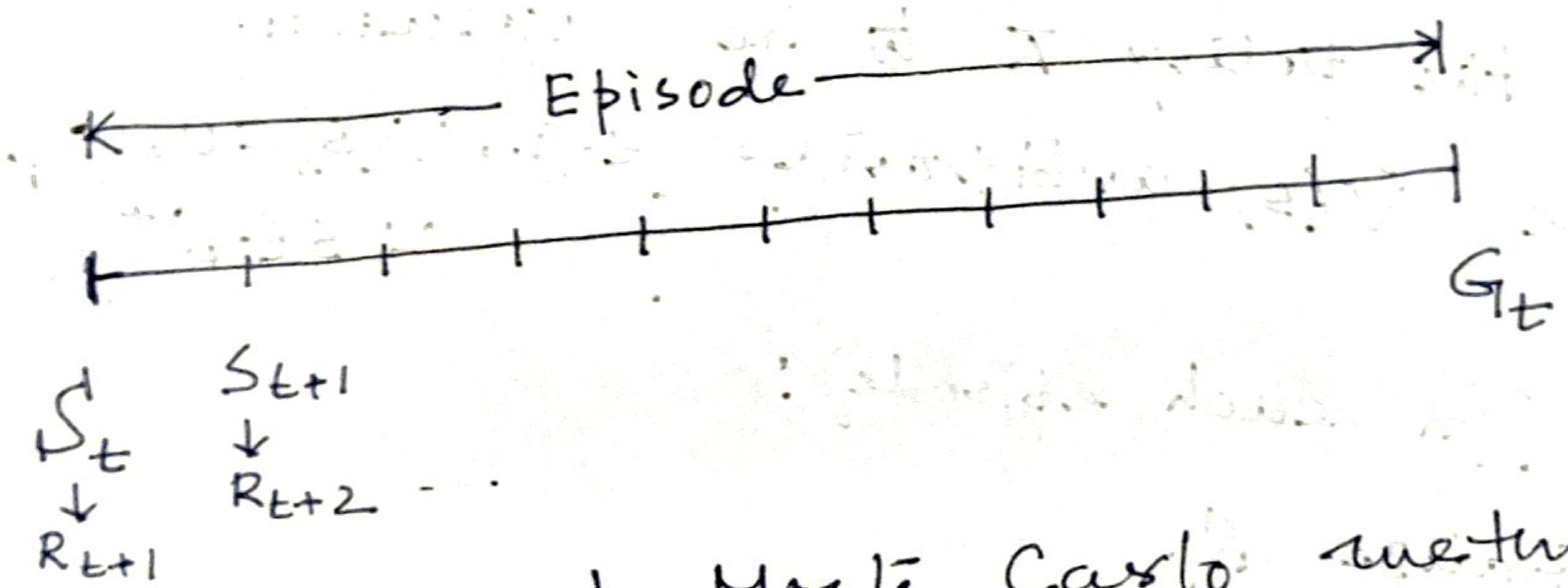
⊛ Example of model free learning:

That is, the transition probabilities/reward/discount factor are not known beforehand. Hence, you can not apply value iteration/policy iteration directly to compute optimal value function/policy function.

* In this chapter, we mainly focus on the policy evaluation/prediction problem, i.e., estimating the value function v_π for a given policy π.

* For the control problem (finding an optimal policy), some variation of generalized policy iteration (GPI) is used

TD Prediction and Monte-Carlo method.



We are intersted in the value function $v^\pi(s)$ where $s = S_t$.

- Both TD and Monte Carlo methods use experience to solve the prediction problem.
- Given some experience following a policy π, both methods update their estimate v of v_π for the non-terminal states S_t occuring in that experience.
- A simple Monte-Carlo method suitable for Non-stationary environments is:

$$V(S_t) \leftarrow V(S_t) + \alpha \left[G_t - V(S_t) \right]$$

Constant-α Monte Carlo Method.

α is step-size parameter.

$G_t \rightarrow$ Actual return following time t.

Monte-Carlo estimate

$$\boxed{v_\pi(s)} = E_\pi[G_t \mid S_t = s]$$

$$= E_\pi[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \ldots \mid S_t = s]$$

Temporal Difference estimate

$$= E_\pi[R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \cdots) \mid S_t = s]$$

$$= E_\pi[R_{t+1} + \gamma V_\pi(S_{t+1}) \mid S_t = s]$$

Tabular TD(0) for estimating v_π:

Input: the policy π to be evaluated.

Initialize $V(s)$ arbitrarily (e.g., $V(s) = 0$, $\forall\, s \in S^+$).

Repeat (for each episode):

- Initialize S.
- Repeat (for each step of episode):
  - $A \leftarrow$ action given by π for S,
  - Take action A; Observe reward, R, and next state, S'.
  - $V(S) \leftarrow V(S) + \alpha[R + \gamma V(S') - V(S)]$
  - $S \leftarrow S'$
- Until S is the terminal state.

• Driving Home example:

- You dive from work to home

- Trying to predict how long it can take.

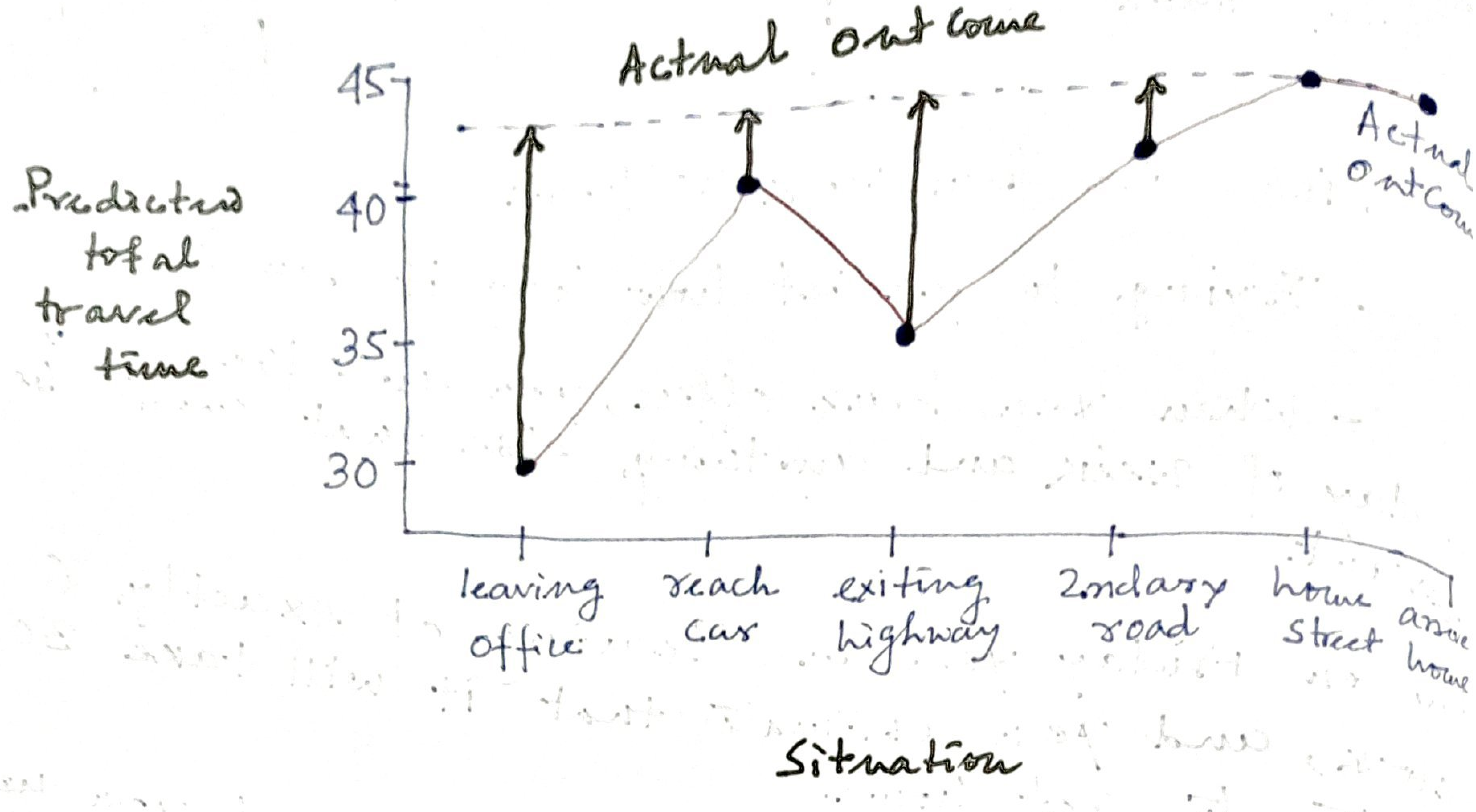
- When you leave office, you note the time, the day of week and anything else that might be relevant.

Say on friday you are leaving at exactly 6 O'clock, and you estimate that it will take 30 minutes to get home.

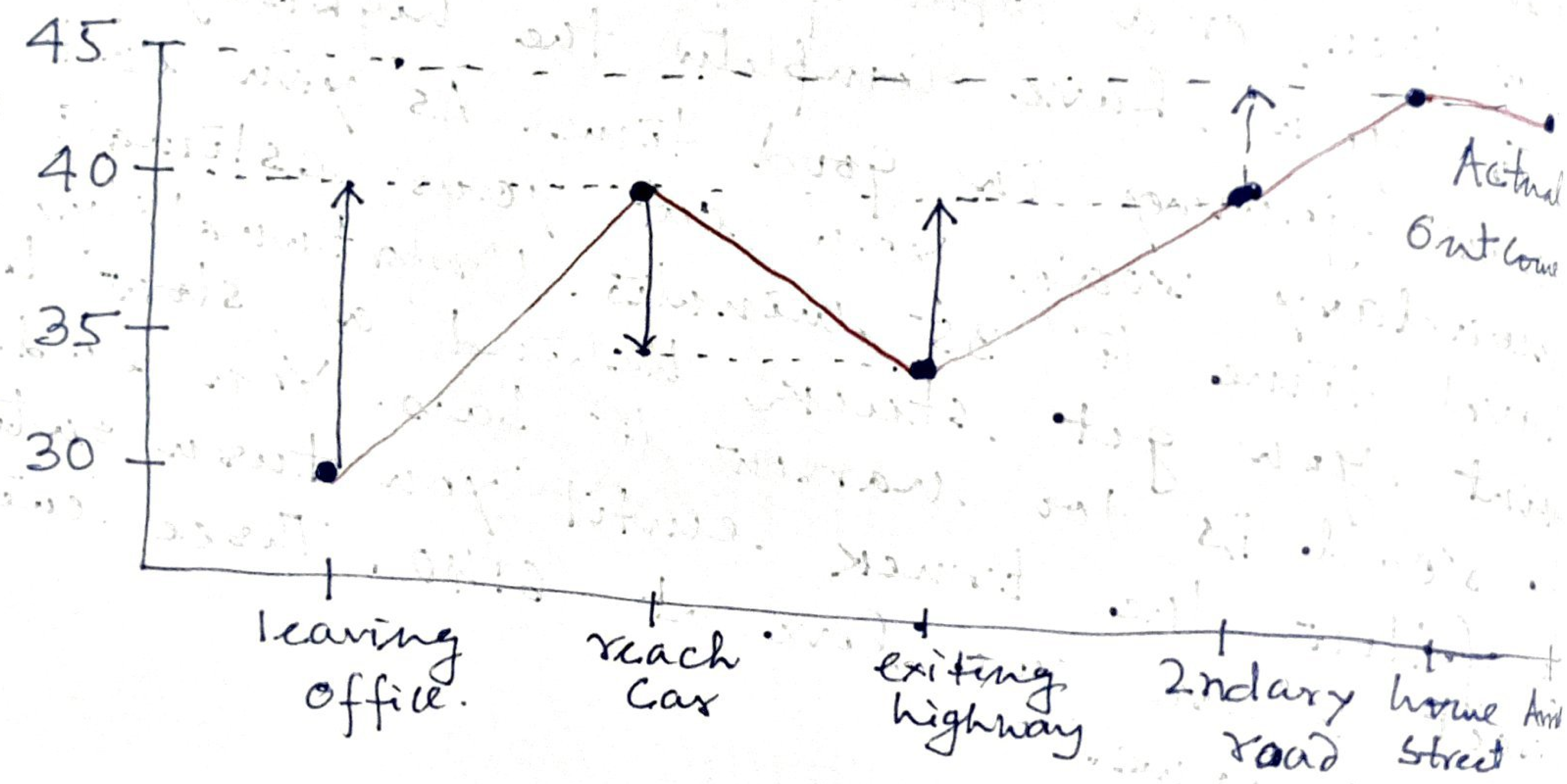
As you reach your car, it is 6:05 and you notice it is starting to rain. Traffic is often slower in rain, so you re-estimate that it will take 35 minutes from then, or a total of 40 minutes. Fifteen minutes later you have completed the highway portion of your journey in good time. As you exit onto a secondary road, you cut your estimate of total travel time to 35 minutes. Unfortunately, at this point you get stuck behind a slow truck, and the road is too narrow to pass. You end up having to follow the truck until you turn onto the side street where you live at 6:40. Three minutes later, you are home.

Formulating the problem:

State	Elapsed Time (minutes)	Predicted Time-to-go.	Predicted Total Time
Leaving office, friday at 6.	0	30	30
Reach car, raining	5	35	40
Exiting highway	20	15	35
2ndary road, behind truck	30	10	40
Entering home street	40	3	43
Arrive home	43	0	43



Changes recommended by Monte Carlo method in the driving home example.



Changes recommended by TD methods in the driving home example.

Next class

- Q-learning + SARSA.
- The Grid World problem.